
BirdGEN: Conditional Spectrogram Augmentation for Bird Species Classification under Simulated Label Scarcity

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Abstract

BirdGEN investigates whether class-conditional spectrogram generation can improve bird-species classification when labeled data are limited. We use recording-isolated log-mel preprocessing, Residual CNN and CRNN classifiers, and a progressively refined conditional VAE. The downstream study preserves nine matched blocks and the four per-species conditions $50 + 0$, $50 + 50$, $50 + 100$, and $50 + 200$. The regenerated VAE pools and reused audited Diffusion pools now support current three-seed CRNN-view quality results, and a recorded equal-count benchmark supports the runtime comparison. In the completed low-resource study, validation selected +200 generated samples per species for both models. Relative to the 84.75% real-only macro-F1 mean, VAE-v3 reached 87.48% and Diffusion reached 86.21%. Quality, classifier utility, and generator-only spectrogram-sampling time are reported as separate outcomes.

Attestation of Teamwork

Shenghao Jin contributed to data preparation and developed and evaluated the conditional VAE. Harvey Li contributed to data preparation, the Residual CNN and CRNN classifiers, the low-resource augmentation experiment, and the VAE-Diffusion generator-only spectrogram-sampling comparison. Vincent Wang developed and evaluated the conditional Diffusion model. All members contributed to the preparation and revision of the final report.

1 Introduction and Problem Formulation

Passive acoustic monitoring enables long-term observation of bird communities, and systems such as BirdNET show that deep learning can convert large audio collections into species-level detections [1]. However, supervised classifiers require sufficiently large and accurately labeled training sets. Ecological audio collections are often imbalanced: common species may have many high-quality recordings, while rare, elusive, or sensitive species have few. Collecting and annotating additional recordings is time-consuming and expensive, and repeated fieldwork may be undesirable in fragile habitats. Limited labeled data therefore remains a major bottleneck for reliable recognition of under-represented species.

Synthetic data augmentation offers a possible way to reduce this bottleneck. ECOGEN represented bird vocalizations as spectrograms, generated additional training examples with a VQ-VAE-based model, and evaluated their value through classification on real recordings [2]. It reported a 12% average improvement in classification accuracy and outperformed conventional augmentation methods

in 80% of the reported comparisons. Inspired by this downstream-oriented evaluation, BIRDCGEN asks whether controllable and computationally affordable class-conditional generators can supplement limited collections of labeled real recordings and improve recognition on unseen real audio. We compare a conditional variational autoencoder (CVAE), which provides tractable encoding and one-pass decoding [3, 4], with a conditional denoising diffusion model. Both generate normalized log-mel spectrograms rather than raw waveforms.

As a controlled proof of concept, we use three common Toronto-area species—American Robin, Northern Cardinal, and Song Sparrow—and artificially restrict the labeled real data available to the downstream classifier. The generators are pretrained on the full training split, whereas each low-resource classifier receives only 50 real spectrograms per species, with or without generated augmentation. Each condition trains a fresh CRNN under matched settings and evaluates it on the same fixed set of real, held-out recordings. Our primary endpoint is

$$\Delta F1 = F1_{\text{real+generated}} - F1_{\text{real-only}},$$

where macro-F1 weights all three species equally. Our secondary question compares VAE and Diffusion in terms of generated-sample quality and the generator-only time required to produce the same number of spectrograms. Reconstruction quality, generated-sample diagnostics, downstream utility, and sampling cost are reported as separate outcomes. This design simulates classifier label scarcity with access to pretrained generators; it does not directly evaluate genuinely rare or unseen species.

2 Data Preparation

The raw recordings were obtained from Vinay Shanbhag’s five-species Bird Song Data Set on Kaggle, which was compiled from Xeno-canto recordings [5]. According to the dataset documentation, the release contains song-only, highest-quality Xeno-canto recordings converted to mono 22.05-kHz WAV files and segmented into three-second clips; its metadata retains the original recording ID, source URL, recordist, and recording-specific license.

We selected American Robin, Northern Cardinal, and Song Sparrow as three target species commonly found in the Toronto area. Our dataset audit verified that all 3,347 metadata-referenced WAV files were present. The resulting subset contains 310 source recording IDs: 1,017 American Robin clips, 1,074 Northern Cardinal clips, and 1,256 Song Sparrow clips. Grouping by recording ID prevents clips from the same recording from crossing train, validation, and test. Historical VAE experiments used 2,339/519/489 clips; the later `content_safe_v2` split removes 24 duplicate training rows and contains 2,315/519/489 clips. Maintained generator comparisons and the refreshed low-resource run use a 510-row generator-safe validation subset, which excludes nine validation rows identified as exact counterparts of historical-training rows. The held-out test set remains unchanged at 489 rows.

Audio is mono at 22.05 kHz and converted to 128×128 log-mel spectrograms using a 1,024-point FFT and hop length 512. The VAE uses training-set global standardization, whereas the maintained classifier maps each clip’s relative 80-dB range to $[-1, 1]$; an adapter converts VAE outputs to the classifier scale.

3 System Design and Models

Figure 1 shows the matched evaluation path. Every downstream condition trains a fresh CRNN and uses the same real test set.

3.1 Classifiers

The 1.66-M-parameter Residual CNN uses six residual blocks and global average/max pooling. It is retained as an architecture-comparison reference, and its frozen features and logits guide VAE-v3 training. It does not score or compare VAE and Diffusion in the maintained independent evaluation. The selected 404-k-parameter CRNN combines convolutional spectral features with a bidirectional GRU and temporal pooling [6]. An independent frozen CRNN supplies generation-quality diagnostics, while freshly initialized CRNNs provide the downstream augmentation evidence.

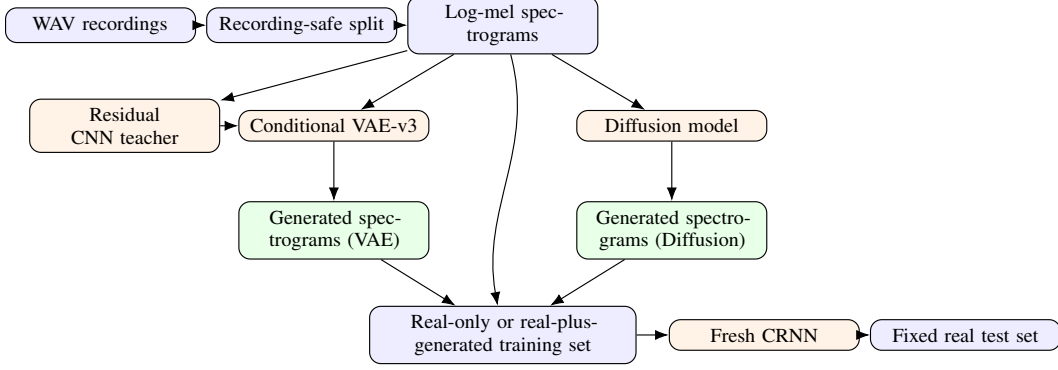


Figure 1: BirdGEN pipeline. Both generated-spectrogram branches feed the same downstream-classifier protocol.

3.2 Conditional VAE Evolution

We selected a conditional VAE (CVAE) rather than an unconditional VAE, because every training spectrogram is paired with a bird-species label, and the downstream task requires controllable, class-labelled augmentation [3, 4]. The species label y conditions both the encoder $q_\phi(z | x, y)$ and decoder $p_\theta(x | z, y)$. Since the requested class is supplied explicitly, the latent variables do not need to represent species identity alone and can thus focus more on within-species variation, such as vocalization type, pitch, and timing. Conditioning reduces cross-species ambiguity, although it does not by itself guarantee realistic or diverse outputs.

V1 established a CVAE baseline with a 128-dimensional global latent vector, species embeddings concatenated at the encoder and decoder, and MSE+KL loss. Its compressed bottleneck and pixel-level reconstruction objective tended to smooth narrow calls, pitch tracks, and transient structure. V2 replaced the global vector with a $16 \times 8 \times 8$ spatial latent and added residual FiLM conditioning [7], resize-convolution, and event-weighted, multiscale, and time/frequency-gradient reconstruction losses. These changes were intended to preserve the locations and edges of short transients and harmonics. For generation, V2 fitted one moment-matched diagonal Gaussian to the training posteriors of each species. Although this reduced the mismatch with standard-normal sampling, a single Gaussian could still average distinct vocalization modes within a species.

V3 increased the latent resolution to $16 \times 16 \times 16$, used a lower KL weight with a longer warmup and per-channel free bits to reduce excessive latent compression and help retain information needed for fine spectro-temporal detail. It also added feature and label consistency losses from a frozen, supervised Residual CNN species classifier trained on labeled real spectrograms, providing species-aware guidance. For generation, V3 replaced each single species-level Gaussian with a train-only posterior-anchor bank, allowing samples to be drawn near multiple encoded vocalization examples. This generation-time strategy mitigates the practical mismatch between the standard-normal prior and learned posteriors, but it does not replace the prior used to train the checkpoint. Since multiple components were modified between versions, the results show the improvement of each complete model rather than the individual causal effect of any single modification.

VAE-v3 has approximately 5.37 M parameters. Species embeddings modulate residual encoder and decoder blocks through FiLM. Its training objective is

$$\mathcal{L} = \mathcal{L}_{\text{event}} + 0.25\mathcal{L}_{\text{ms}} + 0.60(\mathcal{L}_{\Delta t} + \mathcal{L}_{\Delta f}) + \beta\mathcal{L}_{\text{KL,free}} + s_{\text{cls}}[0.08\mathcal{L}_{\text{feat}} + 0.03\mathcal{L}_{\text{label}}]. \quad (1)$$

Here, $\mathcal{L}_{\text{event}}$ is an event-weighted L1 reconstruction term, $\mathcal{L}_{\text{ms}}$ averages L1 losses at multiple resolutions, and $\mathcal{L}_{\Delta t}$ and $\mathcal{L}_{\Delta f}$ are edge-weighted L1 discrepancies between first-order time and frequency differences. The KL coefficient β is warmed up to 5×10^{-4} over 20 epochs, while s_{cls} increases to one over eight epochs. The free-bits term applies a 0.25-nat floor to the batch-and-spatial mean KL of each latent channel, so reducing a channel’s KL below this threshold provides no further optimization benefit. Multiscale and feature-space terms are motivated by [8, 9], and free bits follow [10].

For a training anchor i with the requested species label, the encoder defines a diagonal-Gaussian posterior $q_\phi(z | x_i, y) = \mathcal{N}(\mu_i, \text{diag}(\sigma_i^2))$. The corrected generation-time sampler applies temperature only to its stochastic residual:

$$z = \mu_i + T\sigma_i \odot \epsilon, \quad T = 0.35, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

This correction affects post-training sampling only; the training-time reparameterization and VAE checkpoint were unchanged. The train-only, content-safe posterior bank retained 256 Northern Cardinal, 247 Song Sparrow, and 256 American Robin anchors, and the three generation-seed VAE pools were regenerated from this bank. The method is conceptually related to mixture-of-posteriors priors such as VampPrior [11], but uses fixed encoded training examples rather than learned pseudo-inputs and does not modify the training objective.

3.3 Diffusion Model

4 Experimental Methodology

Classifier architectures were selected on validation data across seeds 42, 123, and 777, then evaluated on held-out test recordings. VAE evidence is separated into paired deterministic reconstruction, unpaired generated-sample quality, and downstream classifier utility. To address the presentation feedback, each frozen generated sample was compared with its nearest same-species test spectrogram in the standardized-logmel domain, with a leave-one-out real-test reference. The test set was used only for this final evaluation; low MSE can also indicate copying or limited diversity.

For the primary experiment, each fresh CRNN sees 50 real examples per species. Three real-subset seeds crossed with three classifier seeds give nine matched blocks. Every block evaluates $50 + 0$, $50 + 50$, $50 + 100$, and $50 + 200$ real-plus-generated samples per species, separately for VAE and Diffusion. Ratios are selected using the 510-row generator-safe validation subset; only the shared $50 + 0$ baseline and each generator’s selected ratio reach the fixed 489-row real test set. Architecture, augmentation, and optimizer-step budget are matched. No existing pretrained classifier is augmented, and the Residual CNN is not a generated-sample evaluator. The recorded runtime benchmark timed five fresh in-memory 600-spectrogram repeats per model on the same GPU, precision, and batch size after warm-up. CUDA was synchronized around generator sampling through the final normalized tensor; model loading, existing arrays, file output, plotting, and metric computation were excluded. Models were measured sequentially: the completed Diffusion benchmark was retained and VAE-v3 was remeasured with the filtered bank, with no Diffusion sampling rerun.

5 Experimental Results

Classifier selection. Among the four evaluated classifier architectures, the 404-k-parameter CRNN achieved the highest controlled three-seed validation macro-F1 ($90.45\% \pm 1.45\%$). The validation-selected seed-777 checkpoint subsequently obtained 89.98% accuracy and 90.16% macro-F1 on the 489-clip held-out real test set.

VAE diagnostics. The recorded test-set reconstruction MSE in the globally standardized log-mel domain was 0.1533 for V1, 0.0654 for V2, and 0.0281 for V3. V2 and V3 used deterministic posterior-mean reconstructions, whereas the historical V1 evaluation sampled from its posterior; the trend is therefore descriptive rather than a strictly matched deterministic comparison. V3 test MAE was 0.1261 and spectral convergence was 0.1794. MSE is an evaluation diagnostic rather than V3’s complete training objective, and these input-to-reconstruction values are not generated-to-real distances.

For the separately verified seed-42 notebook pool (16 samples/species), the mean same-species generated-to-test nearest-neighbor pixel MSE was 0.5120 (SD 0.1998; median 0.4501; IQR 0.3878–0.6231); the leave-one-out real-to-real reference mean was 0.3350 (SD 0.1419; median 0.3185; IQR 0.2368–0.4154). The generated-to-test/real-to-real means by species were 0.4308/0.3237 for American Robin, 0.5579/0.3650 for Northern Cardinal, and 0.5474/0.3175 for Song Sparrow. These unpaired pixel-space summaries use 48 and 489 queries, respectively, and nearest-neighbor MSE can reward copying; they are neither perceptual-realism nor waveform-quality scores. This hash-recorded diagnostic used a separately reconstructed notebook pool (temperature 0.35; train-only posterior

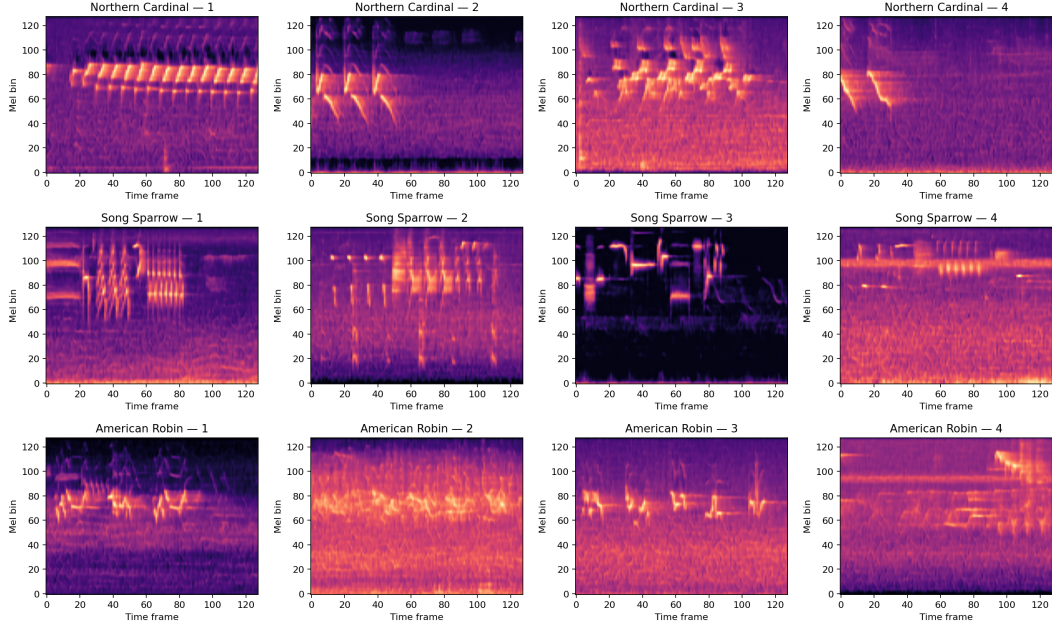


Figure 2: Existing VAE-v3 notebook posterior-anchor samples at temperature 0.35: four examples per species (48 samples were saved in total). Panels are independently color-normalized; spectrogram appearance alone does not establish waveform realism.

Table 1: Frozen selected-CRNN and feature-space generated-sample diagnostics.

Metric	VAE-v3	Diffusion
Target-label accuracy / CRNN macro-F1	96.67% / 96.66%	92.44% / 92.42%
Classifier-feature Fréchet distance	22.8950	41.2550
Feature precision / feature recall	0.8528 / 0.8972	0.5654 / 0.8207
Feature density / coverage / diversity ratio	0.6423 / 0.7419 / 0.8719	0.3114 / 0.4845 / 1.0083
Pixel Wasserstein / near-train fraction / exact copies	0.0787 / 36.67% / 0	0.1916 / 2.50% / 0

bank), rather than the canonical pools used downstream. Although both use standardized log-mels, 0.5120 and 0.0281 are not comparable because they summarize generated-to-nearest-test and input-to-own-reconstruction pairs, respectively. The Residual CNN remains a frozen VAE training teacher and supplies no independent VAE–Diffusion evaluation. Table 1 reports the current audited pools using the selected CRNN and its shared feature space. Values are three-generation-seed means; the feature rows additionally average the three species.

Low-resource augmentation. Test reporting is limited to the shared 50 + 0 baseline and the independently selected +200 condition. Mean held-out macro-F1 was 84.75% for real-only, 87.48% for VAE-v3, and 86.21% for Diffusion. The paired changes were +2.72 percentage points for VAE-v3 (7/9 positive blocks; descriptive block-bootstrap interval +1.00 to +4.65) and +1.46 points for Diffusion (6/9; +0.15 to +2.89).

Equal-count generator timing. The recorded benchmark generated 600 fresh in-memory spectrograms (200 per species) per synchronized repeat after warm-up. The package retains every repeat’s seconds, throughput, peak accelerator memory, hardware, precision, batch size, sampler settings, seed, measurement sequence, and timing boundary. Pre-existing arrays are not timing evidence. The median [Q1, Q3] repeat time was 0.3046 [0.2983, 0.3076] seconds for VAE-v3 and 411.1814 [410.9396, 411.2319] seconds for Diffusion. These measurements use FP32, batch size 8, five warm-up batches, five CUDA-synchronized repeats, and an RTX 4070 SUPER. They exclude loading, conversion, transfer, file I/O, and audio decoding. The completed Diffusion benchmark was retained;

Table 2: Completed low-resource CRNN study across nine matched blocks.

Per-species condition	Val. macro-F1 (VAE / Diff.)	Test status
50 + 0	Shared real-only reference	Evaluated baseline
50 + 50	84.53% / 85.45%	Not selected
50 + 100	85.88% / 86.06%	Not selected
50 + 200	86.47% / 86.84%	Selected for both

Table 3: Generator-only timing comparison for matched fresh outputs.

Model	Outputs/repeat	Repeat seconds	Samples/s	Peak accelerator memory
VAE-v3 (corrected sampler)	600	0.3058 ± 0.0096	1963.276	0.185 GiB
Diffusion (100-step DDIM)	600	412.3090 ± 2.8120	1.455	0.333 GiB
Difference / Diffusion-to-VAE ratio	600	$412.0032 / 1348.09\times$	–	–

VAE-v3 was remeasured afterward using the filtered 759-anchor bank, and no Diffusion sampling was rerun.

6 Discussion

In the simulated low-resource setting, both validation-selected +200 arms improved mean held-out macro-F1 versus their matched real-only controls. The nine blocks reuse overlapping real subsets, so the intervals are descriptive and the largest tested ratio is not an estimated optimum. This result applies to newly initialized low-resource CRNNs, not the historical full-data CRNN. Reconstruction MSE rewards pixel proximity, the Residual CNN is coupled to V3 training and therefore restricted to its teacher role, and a closed-set independent CRNN cannot reject unknown or noisy inputs. Spectrogram and classifier metrics also do not replace listening or waveform-level evaluation [12]. VAE-v3 has higher CRNN target compatibility, lower macro-species Fréchet distance, and higher feature precision/coverage in these pools, but it also has a much larger near-training feature fraction. Diffusion has a diversity ratio nearer one and lower copy-risk flags. Separately, Diffusion took $1348.09\times$ the VAE-v3 generator time under the recorded boundary.

7 Conclusion

BirdGEN now has corrected three-seed generator evidence, a completed matched low-resource CRNN study, and a matched timing comparison. Validation selected +200/species for both models; the mean paired macro-F1 changes were +2.72 points for VAE-v3 and +1.46 points for Diffusion. Quality diagnostics remain metric-specific rather than a composite winner, and the efficiency result is limited to generator-only spectrogram sampling on the recorded hardware.

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A Supplementary Reproducibility Notes

VAE-v3 used AdamW with learning rate 2×10^{-4} , batch size 24, a 20-epoch KL warm-up, and checkpoint selection by the complete validation objective. The portable pool generator now applies the notebook’s sampling temperature 0.35 explicitly. The VAE checkpoint was not retrained. The existing posterior bank was filtered to 256 Northern Cardinal, 247 Song Sparrow, and 256 American Robin anchors, and the three canonical VAE pools were regenerated with the corrected sampler. The reported generated-to-test MSE instead uses a separate, hash-recorded seed-42 pool reconstructed with the original notebook, 16 samples per species, and the train-only posterior bank. This MSE package remains separate from the three-seed augmentation pools and does not replace their quality or downstream CRNN evidence.

For the equal-count timing study, each model generated 600 spectrograms (200 per species) in FP32 with batch size 8 on the same hardware. Five warm-up batches preceded five synchronized repeats. Models were measured sequentially, not interleaved: the completed Diffusion benchmark was retained and VAE-v3 was remeasured after the bank filter. The sole timing boundary is generator sampling through the final normalized spectrogram tensor; model loading, existing arrays, file output, plotting, and metric computation are excluded. The package preserves repeat seconds, throughput, peak accelerator memory, hardware, precision, batch size, seeds, sampler settings, and the timing-boundary description.